esomeprazole (Rx, OTC)

Brand and Other Names: Nexium, Nexium 24HR
Classes: Proton Pump Inhibitors

Dosing & Uses

ADULT

Dosage Forms & Strengths

capsule, delayed-release (as magnesium)

- 20mg (Rx/OTC)
- 40mg
- 24.65mg (strontium salt)
- 49.3mg (strontium salt)

tablet delayed release (as magnesium)

- 20 mg

injection, powder for reconstitution

- 20mg/vial
- 40mg/vial

packets for oral suspension

- 10mg
- 20mg
- 40mg

GERD Without Erosive Esophagitis

20 mg PO qDay for 4 weeks; consider an additional 4 weeks of treatment if symptoms do not resolve completely in the first 4 weeks

GERD With Erosive Esophagitis

20-40 mg PO qDay for 4-8 weeks

If oral therapy inappropriate or not possible: 20-40 mg qDay IV up to 10 days; switch to PO once patient able to swallow

Maintenance: 20 mg PO qDay for up to 6 months

Helicobacter Pylori Eradication

Combination therapy (with amoxicillin and clarithromycin) for eradication of H pylori in patients with duodenal ulcer

40 mg PO qDay for 10 days, PLUS

Amoxicillin 1000 mg PO q12hr for 10 days, PLUS

Clarithromycin 500 mg PO q12hr for 10 days

Risk Reduction of NSAID-Associated Gastric Ulcer

20-40 mg PO qDay for up to 6 months

NSAID-Induced Gastric Ulcer

20 mg PO qDay for 4-8 weeks

Zollinger-Ellison Syndrome

80 mg PO divided q12hr (initial); adjust regimen to efficacy; up to 240 mg PO qDay, OR

120 mg PO q12hr administered to patients

Gastric or Duodenal Ulcers Following Therapeutic Endoscopy

IV indicated for risk reduction of rebleeding of gastric or duodenal ulcers following therapeutic endoscopy in adults

80 mg IV infused over 30 min, THEN continuous IV infusion of 8 mg/hr for total treatment duration of 72 hr

Follow IV therapy with oral acid suppressive therapy

Frequent Heartburn

OTC: 20 mg PO qDay x14 days

Hepatic Impairment

Oral administration

- Mild to moderate (Child-Pugh A/B): No dosage adjustment required
- Severe (Child-Pugh C): Not to exceed 20 mg/day

GERD (IV)

- Mild to moderate (Child-Pugh A/B): No dosage adjustment required
- Severe (Child-Pugh C): Not to exceed 20 mg/day IV

Bleeding gastric or duodenal ulcers (IV)

- No dosage adjustment required with initial 80 mg/30 min IV dose
- Adjust dose for continuous IV infusion
- Mild-to-moderate (Child Pugh A/B): Not to exceed 6 mg/hr
- Severe (Child Pugh C): Not to exceed 4 mg/hr

Eosinophilic Esophagitis (Orphan)

Orphan designation for treatment of eosinophilic esophagitis

Orphan sponsor

- Adare Pharmaceuticals, Inc; 845 Center Drive; Vandalia, Ohio 45377

PEDIATRIC

Dosage Forms & Strengths

capsule, delayed-release

- 20mg
- 40mg

tablet delayed release

- 20mg

injection, powder for reconstitution

- 20mg/vial

- 40mg/vial

packets for oral suspension

- 10mg
- 20mg
- 40mg

GERD Without Erosive Esophagitis

Oral

- <1 year: Safety and efficacy not established
- 1-12 years: 10-20 mg PO qDay for up to 8 weeks
- >12 years: 20-40 mg PO qDay for up to 8 weeks

Short-term Treatment of GERD

IV

- Short-term treatment of GERD with erosive esophagitis when oral therapy is not possible or appropriate
- <1 month: Safety and efficacy not established
- 1 month to 1 year: 0.5 mg/kg IV qDay
- ≥1 year (<55 kg): 10 mg IV qDay
- ≥1 year (≥55 kg): 20 mg IV qDay

GERD With Erosive Esophagitis (Healing)

<1 month: Safety and efficacy not established

1 month to 1 year

- 3.5 kg: 2.5 mg PO qDay for up to 6 weeks
- >3.5-7.5 kg: 5 mg PO qDay for up to 6 weeks
- >7.5 kg: 10 mg PO qDay for up to 6 weeks

1-12 years

- <20 kg: 10 mg PO qDay for 8 weeks
- ≥20 kg: 10-20 mg qDay for 8 weeks

>12 years

- 20-40 mg PO qDay for 4-8 weeks
- Maintenance: 20 mg PO qDay up to 6 months

Interactions

Interaction Checker

Enter a drug name to check for any interactions. Type a drug, OTC, or herbal and esomeprazole
No Interactions Found
Interactions Found

Contraindicated

Serious

Significant - Monitor Closely

Minor

All Interactions Sort By: Severity

Contraindicated (2)

- nelfinavir
  esomeprazole decreases levels of nelfinavir by unspecified interaction mechanism. Contraindicated. Coadministration may lead to loss of nelfinavir virologic response and development of resistance; mechanism may be CYP2C19 inhibition of nelfinavir conversion to active M8 metabolite, and also PPIs decreasing gastric pH resulting in decreased nelfinavir absorption.

- rilpivirine
  esomeprazole decreases levels of rilpivirine by increasing gastric pH. Applies only to oral form of both agents. Contraindicated. Concurrent use may cause treatment failure and/or the development of rilpivirine or NNRTI resistance owing to decreased levels.

Serious (36)

- acalabrutinib
  esomeprazole decreases levels of acalabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Acalabrutinib solubility decreases with increasing gastric pH. Due to the long-lasting effect of PPIs, separation of doses may not eliminate the interaction.

- apalutamide
  apalutamide will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Coadministration of apalutamide, a strong CYP2C19 inducer, with drugs that are CYP2C19 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered.

  apalutamide will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

Coadministration of apalutamide, a strong CYP3A4 inducer, with drugs that are CYP3A4 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered. Adjust dose according to prescribing information if needed.

- atazanavir
  esomeprazole will decrease the level or effect of atazanavir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Atazanavir solubility decreases as pH increases. Substantially reduced plasma concentrations of atazanavir are expected if PPIs are coadministered. PPI dose should not exceed a dose comparable to omeprazole 20 mg and must be taken ~12 h before atazanavir/ritonavir in treatment naive-patients. PPIs are not recommended in treatment-experienced taking atazanavir.

- carbamazepine
  carbamazepine will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- ceritinib
  esomeprazole will decrease the level or effect of ceritinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- cilostazol
  esomeprazole increases toxicity of cilostazol by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Decrease cilostazol dose by 50%; 3,4-dehydrocilostazol, an active metabolite, increased by 69% as a result of omeprazole inhibition of CYP2C19.

- clopidogrel
  esomeprazole decreases effects of clopidogrel by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Clopidogrel efficacy may be reduced by drugs that inhibit CYP2C19. Inhibition of platelet aggregation by clopidogrel is entirely due to an active metabolite. Clopidogrel is metabolized to this active metabolite in part by CYP2C19. .

- dacomitinib
  esomeprazole will increase the level or effect of dacomitinib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Concomitant use with a PPI decreases dacomitinib concentrations, which may reduce dacomitinib efficacy. Avoid use of PPIs with dacomitinib. As an alternative to PPIs, use locally-acting antacids or an H2-receptor antagonist. Administer at least 6 hours before or 10 hours after taking an H2-receptor antagonist.

- dasatinib
  esomeprazole will decrease the level or effect of dasatinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- defactinib
  esomeprazole decreases levels of defactinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Defactinib exposure may be decreased when coadministered with proton pump inhibitors (PPIs). If

concomitant use with PPIs cannot be avoided, separate administration by at least 2 hr.

- digoxin
  esomeprazole will increase the level or effect of digoxin by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- enzalutamide
  enzalutamide will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- erlotinib
  esomeprazole decreases levels of erlotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration of proton pump inhibitors and erlotinib may decrease erlotinib exposure and reduce its efficacy. Consider alternate acid reducing agents.

- fluvoxamine
  fluvoxamine will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- idelalisib
  idelalisib will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Idelalisib is a strong CYP3A inhibitor; avoid coadministration with sensitive CYP3A substrates

- infigratinib (DSC)
  esomeprazole will decrease the level or effect of infigratinib (DSC) by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- isoniazid
  isoniazid will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- itraconazole
  esomeprazole will decrease the level or effect of itraconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- ketoconazole
  esomeprazole will decrease the level or effect of ketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- levoketoconazole
  esomeprazole will decrease the level or effect of levoketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- lonafarnib
  lonafarnib will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Lonafarnib may increase the AUC and peak concentration of CYP2C19 substrates. If

coadministration unavoidable, monitor for adverse reactions and reduce the CYP2C19 substrate dose in accordance with its approved product labeling.

- mesalamine
  esomeprazole decreases effects of mesalamine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Applies only to sustained release dosage form.

- neratinib
  esomeprazole will decrease the level or effect of neratinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- nilotinib
  esomeprazole decreases levels of nilotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. As an alternative to PPIs, use H2-blockers ~10 hr before or ~2 hr after nilotinib, or use antacids ~2 hr before or ~2 hr after nilotinib (does not apply if using ODT nilotinib).

- nisoldipine
  esomeprazole will increase the level or effect of nisoldipine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- pazopanib
  esomeprazole will decrease the level or effect of pazopanib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of pazopanib with drugs that raise gastric pH; consider short-acting antacids in place of PPIs and H2 antagonists; separate antacid and pazopanib dosing by several hours

- pexidartinib
  esomeprazole will decrease the level or effect of pexidartinib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of proton pump inhibitors (PPIs) with pexidartinib. Use H2-receptor antagonists or antacids if needed. When using alternatives to PPIs, administer pexidartinib 2 hr before or after taking locally-acting antacids OR administer pexidartinib at least 2 hr before or 10 hr after taking an H2-receptor antagonist.

- phenobarbital
  phenobarbital will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- rilzabrutinib
  esomeprazole decreases levels of rilzabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease rilzabrutinib exposure, resulting in reduced efficacy.

- secretin
  esomeprazole, secretin. Other (see comment). Avoid or Use Alternate Drug. Comment: Concomitant use of PPIs may cause a hyperresponse in gastrin secretion in response to stimulation testing with secretin, falsely suggesting gastrinoma. The time it takes for serum gastrin concentrations to return to baseline following discontinuation of PPIs is specific to the individual PPI. Temporarily stop

esomeprazole treatment at least 14 days before assessing to allow gastrin levels to return to baseline.

- sofosbuvir/velpatasvir
  esomeprazole will decrease the level or effect of sofosbuvir/velpatasvir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Velpatasvir solubility decreases as gastric pH increases (practically insoluble at pH >5). Coadministration of sofosbuvir/velpatasvir with omeprazole or other PPIs is not recommended. If considered medically necessary, give sofosbuvir/velpatasvir with food 4 hr before omeprazole 20 mg. Use with other PPIs has not been studied.

- sotorasib
  esomeprazole will decrease the level or effect of sotorasib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. If use with an acid-reducing agent cannot be avoided, administer sotorasib 4 hr before or 10 hr after administration of a locally-acting antacid.

- sparsentan
  esomeprazole decreases effects of sparsentan by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Proton pump inhibitors may decrease sparsentan exposure which may reduce efficacy of sparsentan.

- taletrectinib
  esomeprazole decreases levels of taletrectinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease taletrectinib exposure and reduce its efficacy.

- tucatinib
  tucatinib will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Avoid concomitant use of tucatinib with CYP3A substrates, where minimal concentration changes may lead to serious or life-threatening toxicities. If unavoidable, reduce CYP3A substrate dose according to product labeling.

- ziftomenib
  esomeprazole decreases levels of ziftomenib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Proton pump inhibitors decrease zifomenib peak plasma concentations by 70%.

Monitor Closely (84)

- amitriptyline
  esomeprazole will increase the level or effect of amitriptyline by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- amphetamine
  esomeprazole, amphetamine. Other (see comment). Use Caution/Monitor.
  Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- ampicillin
  esomeprazole will decrease the level or effect of ampicillin by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- armodafinil
  armodafinil will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- axitinib
  esomeprazole will increase the level or effect of axitinib by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- belumosudil
  esomeprazole will decrease the level or effect of belumosudil by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Increase belumosudil dosage to 200 mg PO BID when coadministered with proton pump inhibitors.

- belzutifan
  belzutifan will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. If unable to avoid coadministration of belzutifan with sensitive CYP3A4 substrates, consider increasing the sensitive CYP3A4 substrate dose in accordance with its prescribing information.

- bortezomib
  bortezomib will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- bosentan
  bosentan will decrease the level or effect of esomeprazole by increasing metabolism. Use Caution/Monitor.

  bosentan will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- bosutinib
  esomeprazole will decrease the level or effect of bosutinib by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Bosutinib displays pH dependent solubility

- budesonide
  esomeprazole decreases effects of budesonide by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Enteric-coated budesonide dissolves at pH >5.5. Also, dissolution of extended-release budesonide tablets is pH dependent. Coadministration with drugs that increase gastric pH may cause these budesonide products to prematurely dissolve, and possibly affect release properties and absorption of the drug in the duodenum.

- cannabidiol
  esomeprazole will increase the level or effect of cannabidiol by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the cannabidiol dose when coadministered with a moderate CYP2C19 inhibitor.

cannabidiol will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of sensitive CYP2C19 substrates, as clinically appropriate, when coadministered with cannabidiol.

- carbamazepine
  carbamazepine will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- carbonyl iron
  esomeprazole will decrease the level or effect of carbonyl iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cefpodoxime
  esomeprazole will decrease the level or effect of cefpodoxime by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cefuroxime
  esomeprazole will decrease the level or effect of cefuroxime by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cenobamate
  cenobamate will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Increase dose of CYP3A4 substrate, as needed, when coadministered with cenobamate.

  cenobamate will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider a dose reduction of CYP2C19 substrates, as clinically appropriate, when used concomitantly with cenobamate.

- citalopram
  esomeprazole will increase the level or effect of citalopram by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Citalopram 20 mg/day is the maximum recommended dose for patients taking CYP2C19 inhibitors because of the risk of QT prolongation.

- clobazam
  esomeprazole will increase the level or effect of clobazam by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Dosage adjustment may be required; CYP2C19 inhibitors may result in increased exposure to N-desmethylclobazam (active metabolite).

- clozapine
  esomeprazole will decrease the level or effect of clozapine by affecting hepatic enzyme CYP1A2 metabolism. Use Caution/Monitor.

- cobicistat
  cobicistat will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- crizotinib
  esomeprazole decreases levels of crizotinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that elevate the gastric pH may decrease the solubility of crizotinib and subsequently reduce its bioavailability. However, no formal studies have been conducted. .

- crofelemer
  crofelemer increases levels of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Crofelemer has the potential to inhibit CYP3A4 at concentrations expected in the gut; unlikely to inhibit systemically because minimally absorbed.

- cyclosporine
  cyclosporine, esomeprazole. Either increases toxicity of the other by Other (see comment). Use Caution/Monitor. Comment: When used for prolonged periods of time PPIs may cause hypomagnesemia and the risk is further increased when used concomitantly with drugs that also have the same effects.

- dabrafenib
  esomeprazole will decrease the level or effect of dabrafenib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that alter upper GI tract pH (eg, PPIs, H2-blockers, antacids) may decrease dabrafenib solubility and reduce its bioavailability

  dabrafenib will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Use alternative if available

- dextroamphetamine
  esomeprazole, dextroamphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- diazepam buccal
  esomeprazole will increase the level or effect of diazepam buccal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.

- diazepam intranasal
  esomeprazole will increase the level or effect of diazepam intranasal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.

- dichlorphenamide
  dichlorphenamide and esomeprazole both decrease serum potassium. Use Caution/Monitor.

- digoxin
  esomeprazole increases toxicity of digoxin by Other (see comment). Use Caution/Monitor. Comment: Prolonged use of PPIs may cause hypomagnesemia and

increase risk for digoxin toxicity.

- dordaviprone
  dordaviprone and esomeprazole both increase QTc interval. Use Caution/Monitor. Increase the frequency of ECG and electrolyte monitoring.

- duloxetine
  esomeprazole will decrease the level or effect of duloxetine by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- efavirenz
  efavirenz will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- elagolix
  elagolix will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak CYP2C19 inhibitor. Caution with sensitive CYP2C19 substrates.

  elagolix will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak-to-moderate CYP3A4 inducer. Monitor CYP3A substrates if coadministered. Consider increasing CYP3A substrate dose if needed.

- eltrombopag
  esomeprazole will decrease the level or effect of eltrombopag by affecting hepatic enzyme CYP1A2 metabolism. Use Caution/Monitor.

- escitalopram
  esomeprazole will increase the level or effect of escitalopram by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- eslicarbazepine acetate
  eslicarbazepine acetate will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- etravirine
  esomeprazole will increase the level or effect of etravirine by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  etravirine will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- fedratinib
  fedratinib will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Adjust dose of drugs that are CYP3A4 substrates as necessary.

  fedratinib will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Adjust dose of drugs that are CYP2C19 substrates as necessary.

- ferric gluconate

esomeprazole will decrease the level or effect of ferric gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferric maltol
  esomeprazole will decrease the level or effect of ferric maltol by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous fumarate
  esomeprazole will decrease the level or effect of ferrous fumarate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous gluconate
  esomeprazole will decrease the level or effect of ferrous gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous sulfate
  esomeprazole will decrease the level or effect of ferrous sulfate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- fexinidazole
  fexinidazole will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- fluconazole
  fluconazole will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- fosamprenavir
  esomeprazole will decrease the level or effect of fosamprenavir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- fosphenytoin
  esomeprazole will increase the level or effect of fosphenytoin by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  fosphenytoin will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- gefitinib
  esomeprazole decreases levels of gefitinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Avoid coadministration of gefitinib with PPIs if possible. If treatment with a PPI is required, separate gefitinib and PPI doses by 12 hr.

- glipizide
  esomeprazole will increase the level or effect of glipizide by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- glyburide
  esomeprazole will increase the level or effect of glyburide by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- imipramine

esomeprazole will increase the level or effect of imipramine by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- iron dextran complex
  esomeprazole will decrease the level or effect of iron dextran complex by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- iron sucrose
  esomeprazole will decrease the level or effect of iron sucrose by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- istradefylline
  istradefylline will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Istradefylline 40 mg/day increased peak levels and AUC of CYP3A4 substrates in clinical trials. This effect was not observed with istradefylline 20 mg/day. Consider dose reduction of sensitive CYP3A4 substrates.

- ledipasvir/sofosbuvir
  esomeprazole decreases levels of ledipasvir/sofosbuvir by Other (see comment). Use Caution/Monitor. Comment: Ledipasvir solubility decreases as pH increases; drugs that increase gastric pH are expected to decrease levels of ledipasvir; proton-pump inhibitor doses comparable to omeprazole <20 mg can be administered simultaneously with ledipasvir/sofosbuvir under fasted conditions.

- lenacapavir
  lenacapavir will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Lencapavir (a moderate CYP3A4 inhibitor) may increase CYP3A4 substrates initiated within 9 months after last SC dose of lenacapavir, which may increase potential risk of adverse reactions of CYP3A4 substrates.

- letermovir
  letermovir increases levels of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- lumacaftor/ivacaftor
  lumacaftor/ivacaftor, esomeprazole. affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. In vitro studies suggest that lumacaftor may induce and ivacaftor may inhibit CYP2C19 substrates. .

  lumacaftor/ivacaftor will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Lumacaftor/ivacaftor is a strong CYP3A4 inhibitor and also has the potential to induce CYP2C19 and both induce and inhibitor P-gp.

- mavacamten
  esomeprazole increases levels of mavacamten by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Modify mavacamten dose if coadministered with weak or moderate CYP2C19 inhibitors.

- methotrexate

esomeprazole increases levels of methotrexate by decreasing renal clearance. Use Caution/Monitor. Increased risk of toxicity with higher doses.

- methylphenidate
  esomeprazole decreases effects of methylphenidate by enhancing GI absorption. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Since the characteristics of methylphenidate extended release capsules (Ritalin LA) are pH dependent, coadministration of antacids or acid suppressants could alter the release of methylphenidate. Consider separating the administration of the antacid and the methylphenidate extended-release capsules may be avoided.

- mitapivat
  mitapivat decreases levels of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Monitor patients for loss of therapeutic effect if taking sensitive CYP2C19 substrates. .

- mitotane
  mitotane decreases levels of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Mitotane is a strong inducer of cytochrome P-4503A4; monitor when coadministered with CYP3A4 substrates for possible dosage adjustments.

- mycophenolate
  esomeprazole will decrease the level or effect of mycophenolate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Potential interaction applies to mycophenolate mofetil. Enteric coated mycophenolate sodium formulation is less sensitive to this interaction. Clinical significance unclear.

- paltusotine
  esomeprazole decreases levels of paltusotine by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Coadministration demonstrated a dose-dependent decrease in paltusotine exposure owing to decreased paltusotine solubility at elevated gastric pH. May require increased paltusotine dose. Avoid coadministration if patient is already taking paltusotine 60 mg.

- pentobarbital
  pentobarbital will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- phenobarbital
  phenobarbital will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- phenytoin
  esomeprazole will increase the level or effect of phenytoin by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  phenytoin will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- polysaccharide iron

esomeprazole will decrease the level or effect of polysaccharide iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- posaconazole
  esomeprazole will decrease the level or effect of posaconazole by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

  posaconazole will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- primidone
  primidone will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- rifampin
  rifampin will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

  rifampin will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- riociguat
  esomeprazole will decrease the level or effect of riociguat by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ritonavir
  ritonavir will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- rose hips
  esomeprazole will decrease the level or effect of rose hips by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- sparsentan
  sparsentan will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Sparsentan (a CYP2C19 inducer) decreases exposure of CYP2C19 substrates and reduces efficacy related to these substrates.

- St John's Wort
  St John's Wort will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  St John's Wort will decrease the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- stiripentol
  stiripentol will decrease the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of CYP2C19 substrates, if adverse reactions are experienced when administered concomitantly with stiripentol.

- tacrolimus

esomeprazole will increase the level or effect of tacrolimus by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Concomitant administration may increase tacrolimus whole blood concentrations, particularly in intermediate or poor metabolizers of CYP2C19

- taletrectinib
  taletrectinib and esomeprazole both increase QTc interval. Use Caution/Monitor. Concomitant use may increase the risk of QTc interval prolongation. If concomitant use is unavoidable, adjust the frequency of ECG and electrolyte monitoring.

- triclabendazole
  triclabendazole will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. If plasma concentrations of the CYP2C19 substrates are elevated during triclabendazole, recheck plasma concentration of the CYP2C19 substrates after discontinuation of triclabendazole.

- voriconazole
  voriconazole will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  voriconazole will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- ziftomenib
  esomeprazole, ziftomenib. QTc interval. Use Caution/Monitor. Caution if ziftomenib is coadministered with drugs that exacerbated risk of QT prolongation (eg, hypokalemia, orthostatic hypotension, inhibits metabolism of ziftomenib).

Minor (46)

- acetazolamide
  acetazolamide will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- alosetron
  esomeprazole will decrease the level or effect of alosetron by increasing metabolism. Minor/Significance Unknown.

- alprazolam
  esomeprazole increases levels of alprazolam by decreasing metabolism. Minor/Significance Unknown.

- ambrisentan
  esomeprazole will increase the level or effect of ambrisentan by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- anastrozole
  anastrozole will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- bismuth subsalicylate

esomeprazole increases levels of bismuth subsalicylate by enhancing GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

- blessed thistle
  blessed thistle decreases effects of esomeprazole by pharmacodynamic antagonism. Minor/Significance Unknown. Theoretical interaction.

- bosentan
  esomeprazole will increase the level or effect of bosentan by affecting hepatic enzyme CYP2C9/10 metabolism. Minor/Significance Unknown.

- capecitabine
  esomeprazole decreases effects of capecitabine by unknown mechanism. Minor/Significance Unknown. Retrospective data suggest elevated gastric pH caused by PPI use may impair capecitabine tablet dissolution and/or reduce absorption. However, a randomized crossover study found esomeprazole did not affect capecitabine systemic exposure.

- chlordiazepoxide
  esomeprazole will increase the level or effect of chlordiazepoxide by decreasing metabolism. Minor/Significance Unknown.

- clomipramine
  esomeprazole will increase the level or effect of clomipramine by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- clonazepam
  esomeprazole will increase the level or effect of clonazepam by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- cyanocobalamin
  esomeprazole decreases levels of cyanocobalamin by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

- cyclophosphamide
  cyclophosphamide will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- danazol
  danazol will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- darifenacin
  darifenacin will decrease the level or effect of esomeprazole by Other (see comment). Minor/Significance Unknown. Effectiveness of proton pump inhibitors may be decreased theoretically if administered with other antisecretory agents

- devil's claw
  devil's claw decreases effects of esomeprazole by pharmacodynamic antagonism. Minor/Significance Unknown.

- dexamethasone

dexamethasone will decrease the level or effect of esomeprazole by increasing metabolism. Minor/Significance Unknown.

- diazepam
  esomeprazole will increase the level or effect of diazepam by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- dronedarone
  dronedarone will increase the level or effect of esomeprazole by decreasing metabolism. Minor/Significance Unknown.

- drospirenone
  drospirenone will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- efavirenz
  efavirenz will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- elvitegravir/cobicistat/emtricitabine/tenofovir DF
  elvitegravir/cobicistat/emtricitabine/tenofovir DF, esomeprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown. Based on drug interaction studies conducted with the components of Stribild, no clinically significant drug interactions have been either observed or are expected when coadministered with PPIs.

- eslicarbazepine acetate
  eslicarbazepine acetate decreases levels of esomeprazole by increasing metabolism. Minor/Significance Unknown. Monitor for GI symptoms; net increased or decreased effect on PPI action unclear due to opposing CYP450 actions.

- estazolam
  esomeprazole will increase the level or effect of estazolam by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- ferric carboxymaltose
  esomeprazole will decrease the level or effect of ferric carboxymaltose by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.

- fluoxetine
  fluoxetine will increase the level or effect of esomeprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- flurazepam
  esomeprazole increases levels of flurazepam by decreasing metabolism. Minor/Significance Unknown.

- fluvastatin
  esomeprazole will increase the level or effect of fluvastatin by affecting hepatic enzyme CYP2C9/10 metabolism. Minor/Significance Unknown.

- itraconazole

itraconazole will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- larotrectinib
  larotrectinib will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- levoketoconazole
  levoketoconazole will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- levothyroxine
  esomeprazole decreases levels of levothyroxine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- liothyronine
  esomeprazole decreases levels of liothyronine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- liotrix
  esomeprazole decreases levels of liotrix by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- lisdexamfetamine
  esomeprazole, lisdexamfetamine. Other (see comment). Minor/Significance Unknown. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- lorazepam
  esomeprazole increases levels of lorazepam by decreasing metabolism. Minor/Significance Unknown.

- methamphetamine
  esomeprazole decreases levels of methamphetamine by Other (see comment). Minor/Significance Unknown. Comment: Time to maximum concentration (Tmax) of amphetamine is decreased compared to when administered alone; monitor patients for changes in clinical effect and adjust therapy based on clinical response.

- midazolam
  esomeprazole increases levels of midazolam by decreasing metabolism. Minor/Significance Unknown.

- phytoestrogens
  esomeprazole decreases levels of phytoestrogens by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

- ponatinib
  esomeprazole decreases levels of ponatinib by increasing gastric pH. Applies only

to oral form of both agents. Minor/Significance Unknown.

- ribociclib
  ribociclib will increase the level or effect of esomeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- theophylline
  esomeprazole increases toxicity of theophylline by Other (see comment). Minor/Significance Unknown. Comment: Prolonged use of proton pump inhibitors can cause hypochlorhydria, which in turn causes peristalsis in small intestine to increase and peristalsis in the proximal colon to decrease; monitor for toxicity.

- thyroid desiccated
  esomeprazole decreases levels of thyroid desiccated by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- triazolam
  esomeprazole increases levels of triazolam by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- voriconazole
  esomeprazole will increase the level or effect of voriconazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

Adverse Effects

>10%

Headache (2-11%)

1-10%

Flatulence (10%)

Indigestion (6%)

Nausea (6%)

Abdominal pain (1-6%)

Diarrhea (2-4%)

Xerostomia (3-4%)

Dizziness (2-3%)

Constipation (2-3%)

Somnolence (1-2%)

Pruritus (1%)

<1%

Blood and lymphatic system disorders: Agranulocytosis, pancytopenia

Blurred vision

GI disorders: Pancreatitis, stomatitis, microscopic colitis

Hepatobiliary disorders: Hepatic failure, hepatitis with or without jaundice

Anaphylactic reaction/shock

GI candidiasis

Hypomagnesemia

Musculoskeletal disorders: Muscular weakness, myalgia, bone fracture

Nervous system disorders: Hepatic encephalopathy, taste disturbance

Psychiatric disorders: Aggression, agitation, depression, hallucination

Interstitial nephritis

Gynecomastia

Bronchospasm

Skin and subcutaneous tissue disorders: Alopecia, erythema multiforme, hyperhidrosis, photosensitivity, Stevens-Johnson syndrome, toxic epidermal necrolysis (sometimes fatal fatal)

Postmarketing Reports

Cutaneous and systemic lupus erythematosus

Cyanocobalamin (vitamin B-12) deficiency

Clostridium difficile-associated diarrhea

Fundic gland polyps

Acute tubulointerstitial nephritis

Hypocalcemia

Drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalized exanthematous pustulosis (AGEP)

Erectile dysfunction

Warnings

Contraindications

Hypersensitivity to esomeprazole or other proton pump inhibitors (PPIs)

Patients receiving rilpivirine- containing products

Cautions

Do not use if allergic to the drug; may cause severe skin reactions; symptoms may include, skin reddening, blisters, rash; if allergic reaction occurs, discontinue use and seek medical help immediately

PPIs are possibly associated with increased incidence of Clostridium difficile-associated diarrhea (CDAD); consider diagnosis of CDAD for patients taking PPIs who have diarrhea that does not improve

PPIs may decrease the efficacy of clopidogrel by reducing the formation of the active metabolite

Severe hepatic impairment

Cutaneous lupus erythematosus (CLE) and systemic lupus erythematosus (SLE) reported with PPIs; avoid using for longer than medically indicated; discontinue if signs or symptoms consistent with CLE or SLE are observed and refer patient to specialist; most patients improve with discontinuation of PPI alone in 4-12 weeks; serological testing (e.g. ANA) may be positive and elevated serological test results may take longer to resolve than clinical manifestations

Severe cutaneous adverse reactions, including Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalized exanthematous pustulosis (AGEP) reported in association with use of PPIs; discontinue therapy at first signs or symptoms of severe cutaneous adverse reactions or other signs of hypersensitivity and consider further evaluation

Relief of symptoms does not eliminate the possibility of a gastric malignancy; consider additional follow-up and diagnostic testing in adult patients who have suboptimal response or early symptomatic relapse after completing treatment with a PPI

Therapy increases risk of Salmonella, Campylobacter, and other infections

Contains enteric coated granules (acid labile); do not chew or crush; take 1 hr before meals

Published observational studies suggest that PPI therapy may be associated with an increased risk for osteoporosis-related fractures of the hip, wrist, or spine; particularly with prolonged (>1 yr), high-dose therapy

Decreased gastric acidity increases serum chromogranin A (CgA) levels and may cause false-positive diagnostic results for neuroendocrine tumors; temporarily discontinue PPIs before assessing CgA levels

Daily long-term use (e.g., longer than 3 years) may lead to malabsorption or a deficiency of cyanocobalamin

If patient taking a prescription drug, the patient should ask a doctor or a pharmacist whether acid reducers can be taken concomitantly with it

Acute interstitial nephritis reported in patients taking proton pump inhibitors

May elevate and/or prolong serum concentrations of methotrexate and/or its metabolite when administered concomitantly with PPIs, possibly leading to toxicity; consider a temporary withdrawal of PPI therapy with high dose methotrexate administration

Stop use and inform a healthcare professional if rash or joint pain develops

PPI therapy is associated with increased risk of fundic gland polyp; risk increases with long-term use >1 year; patient may be asymptomatic; problem usually identified incidentally on endoscopy; use shortest duration of therapy appropriate to condition being treated

Acute tubulointerstitial nephritis (TIN) reported in patients taking PPIs; may occur at any point during PPI therapy; patients may present with varying signs and symptoms from symptomatic hypersensitivity reactions to non-specific symptoms of decreased renal function (e.g., malaise, nausea, anorexia); in reported case series, some patients were diagnosed on biopsy and in absence of extra-renal manifestations (eg, fever, rash or arthralgia); discontinue therapy and evaluate patients with suspected acute TIN

Hypomagnesemia and mineral metabolism

- Hypomagnesemia may occur with prolonged use (ie, >1 yr; adverse effects may result and include tetany, arrhythmias, and seizures; in 25% of cases reviewed, magnesium supplementation alone did not improve low serum magnesium levels, and the PPI had to be discontinued; consider monitoring magnesium levels prior to initiation of PPI treatment and periodically
- Hypomagnesemia may lead to hypocalcemia and/or hypokalemia and may exacerbate underlying hypocalcemia in at-risk patients; in most patients, treatment of hypomagnesemia required magnesium replacement and discontinuation of the PPI
- Consider monitoring magnesium and calcium levels prior to initiation of therapy and periodically while on treatment in patients with a preexisting risk of hypocalcemia (eg, hypoparathyroidism); supplement with magnesium and/or calcium as necessary. If hypocalcemia is refractory to treatment, consider discontinuing the PPI

Pregnancy & Lactation

Pregnancy

There are no adequate and well-controlled studies in pregnant women; esomeprazole is the S-isomer of omeprazole; available epidemiologic data fail to demonstrate an increased risk of major congenital malformations or other adverse pregnancy outcomes with first trimester omeprazole use; reproduction studies in rats and rabbits resulted in dose-dependent embryo-lethality at omeprazole doses that were approximately 3.4 to 34 times an oral human dose of 40 mg (based on a body surface area for a 60 kg person)

Lactation

Esomeprazole is the S-isomer of omeprazole and limited data suggest that omeprazole may be present in human milk; there are no clinical data on effects of esomeprazole on breastfed infant or on milk production; developmental and health benefits of breastfeeding should be considered along with mother’s clinical need for therapy and any potential adverse effects on breastfed infant from treatment or from underlying maternal condition

Pregnancy Categories

A: Generally acceptable. Controlled studies in pregnant women show no evidence of fetal risk.

B: May be acceptable. Either animal studies show no risk but human studies not available or animal studies showed minor risks and human studies done and showed no risk.

C: Use with caution if benefits outweigh risks. Animal studies show risk and human studies not available or neither animal nor human studies done.

D: Use in LIFE-THREATENING emergencies when no safer drug available. Positive evidence of human fetal risk.

X: Do not use in pregnancy. Risks involved outweigh potential benefits. Safer alternatives exist.

NA: Information not available.

Pharmacology

Mechanism of Action

S-isomer of omeprazole; PPI; binds to H+/K+-exchanging ATPase (proton pump) in gastric parietal cells, resulting in blockage of acid secretion

Absorption

Bioavailability: PO: 89-90%; food decreases AUC by 33-53%; take 1 hr before meal

Onset: 1-2 hr (gastric acid inhibition); within 4 wk (GERD)

Duration (multiple dose): Gastric acid inhibition; PO: 17 hr

Peak plasma time: PO: 1-1.6 hr

Distribution

Protein bound: 97%

Vd: 16 L

Metabolism

Liver; extensively metabolized by hepatic P450 enzyme; major metabolic pathway is via CYP2C19; the rest is via CYP3A4

Metabolites: 5-hydroxyesomeprazole (inactive), esomeprazole sulfone (inactive), desmethyl-esomeprazole (activity unknown)

Enzymes inhibited: CYP2C19

Slow metabolizers (3% of Caucasians and African-Americans) are deficient in CPY2C19 enzyme system; plasma concentration can be higher than in persons who have the enzyme

Elimination

Half-life: 1.2-1.5 hr

Total body clearance: 9-16 L/hr

Excretion: Urine (80%); feces (20%)

Pharmacogenomics

Metabolites of esomeprazole lack antisecretory activity

The major part of esomeprazole's metabolism is dependent on the CYP2C19 isoenzyme, which forms the hydroxy and desmethyl metabolites

CYP2C19 isoenzyme exhibits polymorphism in the metabolism of esomeprazole, since some 3% of Caucasians and 15-20% of Asians lack CYP2C19 and are termed poor metabolizers

At steady state, the ratio of AUC in poor metabolizers to AUC in the rest of the population (extensive metabolizers) is approximately 2

Administration

Oral Administration

If unable to swallow capsule whole, capsule can be opened, emptied on applesauce, mixed, and swallowed immediately

IV Compatibilities

Solution: NS, LR, D5W

IV Preparation

IV injection

- Reconstitute contents of 1 vial with 5 mL NS
- Store at room temp (no refrigeration required) and administer within 12 hr

Intermittent IV infusion

- Reconstitute 1 vial with 5 mL NS, LR, or D5W
- Further dilute to 50 mL
- Store at room temp (no refrigeration required) and administer within 12 hr if diluted in NS or LR, and within 6 hr if diluted in D5W

Continuous IV infusion

- Reconstitute two 40 mg vials with 5 mL each of 0.9% NaCl
- Further dilute the 2 reconstituted vials in 100 mL 0.9% NaCl

IV Administration

IV injection

- Injection: Over no less than 3 min

Intermittent IV infusion

- Infuse over 10-30 min regardless of amount
- Flush IV line with NS, LR, or D5W prior to and after administration
- Do not administer with any other drugs

Continuous IV infusion

- Administer initial 80 mg IV dose over 30 min, THEN follow with
- Continuous IV infusion of 8 mg/hr for total treatment duration of 72 hr

Formulary

Adding plans allows you to compare formulary status to other drugs in the same class.

Formulary information is available for health plans only in the United States.